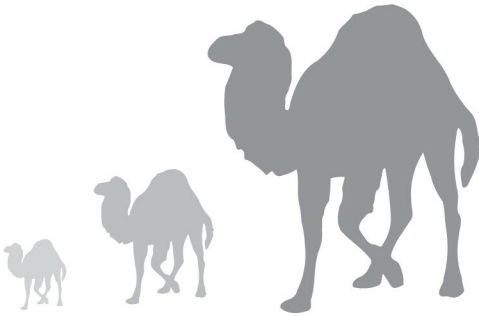


Regular Expressions in Programming Languages: Perls of Wisdom

This article is a continuation of our series on regular expressions, and discusses regular expressions in Perl.



Perl is a general-purpose, interpreted programming language that is most suitable for text processing. Like any other programming language that is mainly used for text processing, Perl also has its own powerful repertoire of regular expressions. In fact, it is a trendsetter in this regard. The regular expression style used in Perl is so well known that it is often called Perl-style regular expressions. It is even adopted by many other popular programming languages, and some even advertise that their regular expressions are as powerful as Perl regular expressions. As mentioned in the previous article in this series, even Python uses Perl-style regular expressions with the help of a regular expression engine called PCRE (Perl Compatible Regular Expressions).

Perl 5 or Perl 6?

Perl was developed by Larry Wall in 1987. Perl is not an acronym but nowadays the backronym Practical Extraction and Report Language is gaining popularity. Perl 5 and Perl 6 are the two active versions of Perl currently being developed and maintained. The command `perl -v` executed on the terminal will tell you the version installed in your system. Most of the popular Linux distributions have Perl 5 as the default option. The only major complete implementation of Perl 6 is called the Rakudo compiler. So, if you want Perl 6 installed in your system, use the Rakudo Perl compiler.

In the beginning of this article, I have called Perl an

interpreted language and now I say that Perl 6 has a compiler. I am sure two things are clear from this. First, Perl 5 and Perl 6 have some major differences between them. Second, people still mean Perl 5 and not Perl 6 when they say Perl. If this article was about the programming language Perl in general, then I would have explained a lot of differences between Perl 5 and Perl 6. But in this article, I am only interested in regular expressions in Perl and nothing more. Unlike Python 2 and Python 3, where the differences in regular expressions are minimal or negligible, Perl 5 and Perl 6 regular expressions have some major differences.

After some consideration, I have decided to stick with Perl 5 and its regular expressions. I would like to point out the reasons for this choice. First, Perl 5 is still the widely accepted Perl standard. Perl 6 might be the future of Perl but compatibility with Perl 5 is not a goal for Perl 6. In fact, some of the Perl 5 fans even complain that Perl 6 is not even real Perl. The two are so different that the Perl camel is no longer the mascot of Perl 6; instead, it is Camelia the butterfly for Perl 6. Also remember that the Perl Foundation supports the onion symbol as the mascot of Perl 5, not the camel. Figure 1 shows the camel of Perl 5 propagated by the O'Reilly media, the onion symbol supported by the Perl Foundation and the butterfly of Perl 6.

So, if I centre my discussions around Perl 6, many of the Perl 5 developers who are far more in number, may not like the code being discussed. This might be the practical reason for sidestepping Perl 6. But there is a theoretical reason also for